

Personalized Profiling of Human Resilience to Heat and Cold Using Machine Learning

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Abstract

Individual responses to heat and cold vary substantially, and resilience to one form of thermal stress does not necessarily imply resilience to the other. This study presents a personalized machine-learning framework in which heat and cold resilience are represented as separate but complementary dimensions of an individual temperature-resilience profile. The analysis used a reference dataset of 2,000 profiles described by 205 physiological, biochemical, individual, and environmental features. These data were reduced to 14 physiologically interpretable components - seven related to heat response and seven to cold response - and summarized into a Heat Resilience Index (H) and a Cold Resilience Index (C). The resulting two-dimensional representation defined four profiles: increased resilience to heat and cold (3%), predominantly cold-resilient (12%), predominantly heat-resilient (26%), and reduced resilience to heat and cold (59%). Machine-learning models were used to assess profile separability and to determine how much information was required for reliable classification. Among three class-balanced Extra Trees models, the two-index H-C model achieved the highest performance (accuracy 98.45%; macro F1 97.91%), while the full post-protocol model also achieved high classification quality; baseline and early-response features alone were insufficient for reliable identification of the final profile. The proposed framework provides a compact, interpretable representation of individual responses to high and low temperatures and may support future occupational assessment in settings characterized by heat, cold, or pronounced temperature contrasts. Potential applications include Arctic industries as well as hot-climate environments and cold-chain operations. Independent validation in relevant occupational cohorts is required before practical use.

Keywords: machine learning; personalized assessment; temperature resilience; heat stress; cold stress; occupational assessment; human physiology

1. Introduction

1.1. Individual variability in responses to heat and cold

Human responses to thermal stress show substantial inter-individual variability even under comparable environmental conditions. The magnitude and pattern of thermoregulatory responses depend not only on the intensity of heat or cold exposure, but also on individual characteristics such as body composition and morphology, age, aerobic fitness, acclimatization status, and other physiological factors. Marked between-person differences have been reported both in heat exchange during exposure to high temperatures and in thermoregulatory responses to cold [Notley et al., 2019; Cramer et al., 2022; Haman et al., 2022]. Heat and cold resilience may also be asymmetric: high resilience to one type of exposure does not necessarily imply high resilience to the other [Park et al., 2010; Kim et al., 2016]. This supports viewing the response to thermal stress as a multidimensional individual characteristic that emerges from the interaction between the person and the environment. Consequently, a single physiological marker may be insufficient to characterize an individual's overall temperature resilience.

In occupational settings, temperature resilience is relevant not only to preventing adverse health effects of heat or cold exposure, but also to maintaining functional capacity during sustained thermal stress. High temperatures can reduce physical work capacity, whereas cold can impair manual performance and other functions required for effective work [Smallcombe et al., 2022; Ray et al., 2019]. Personalized assessment of thermal responses should therefore consider the relationship between temperature resilience and the preservation of functional capacity in the relevant working environment. This provides a rationale for moving beyond generalized assessment toward individualized characterization of responses to heat and cold [Buller et al., 2018; Kingma et al., 2021].

1.2. From biomarker-based assessment of adaptive capacity to multidimensional modeling

A previous study [Zakharinskiy, 2024] examined individual adaptive capacity under professionally oriented examination stress. Participants were aviation students of comparable age enrolled at the same university and living under similar everyday conditions. Physiological measurements were obtained before and after stress exposure during an examination week, using salivary bioluminescence characteristics and cortisol levels as biomarkers.

The data allowed participants to be grouped according to differences in physiological response and adaptive capacity. The study showed that even under similar external conditions, the ability to cope with a stressor can vary substantially across individuals.

The present work extends this line of personalized assessment to a different problem: human resilience to high and low temperatures. Rather than relying on a limited set of biomarkers, the current approach uses a multidimensional representation of the individual response to thermal stress.

1.3. Limitations of existing approaches

Contemporary personalized medicine and deep-phenotyping approaches integrate physiological, biochemical, behavioral, and other individual-level data to construct personalized models of human health and function [Shilo et al., 2020; Reicher et al., 2025].

In thermal-exposure research, computational methods have been applied to thermal comfort assessment, prediction of individual physiological variables, and analysis of responses to high or low temperatures [Jung et al., 2019; Buller et al., 2018; Dolson et al., 2022; Kong et al., 2025]. Other studies have examined heat and cold responses within the same participant cohorts, ranging from physiological comparisons of heat and cold tolerance to four-category classifications based on subjective thermal tolerance and machine-learning approaches to individualized heat and cold comfort [Park et al., 2010; Kim et al., 2016; Zhao et al., 2026].

However, these approaches are largely focused on specific physiological mechanisms, subjective tolerance, or thermal comfort. A compact personalized profile that quantitatively represents resilience to high and low temperatures as two separate dimensions, while integrating physiological, biochemical, individual, and environmental information, remains insufficiently developed.

1.4. Study objective and contribution

The objective of this study was to develop a personalized quantitative approach to heat and cold resilience in which the two responses are represented separately within a single individual profile.

The main contribution is a two-dimensional model of temperature resilience in which 205 analytical features are reduced to 14 physiologically interpretable components and two integrated indices: the Heat Resilience Index (H) and the Cold Resilience Index (C). Their combination forms four individual temperature-resilience profiles. Machine-learning methods were used to analyze the separability of these profiles in the multidimensional feature space and to determine how much information is required for reliable profile classification.

2. Materials and Methods

2.1. Data and cohort construction

The study used a reference cohort of 2,000 individual profiles based on distributions and dependencies reported in open physiological and population data sources, including NHANES [NCHS, 2024], PROSPIE [Havenith et al., 2024a], HEATSHIELD [Havenith et al., 2024b], Cold Pressor OSF [Goreis et al., 2025], and PhysioNet [Goldberger et al., 2000], together with published experimental data on cortisol responses [Silva et al., 2019].

Participants from the source studies were not pooled directly; the available data were used to calibrate the cohort parameters. The cohort was balanced by sex, comprising 1,000 female and 1,000 male profiles, and included baseline physiological state, two heat sessions, three cold-exposure trials, recovery measures, and biochemical response variables.

The final analytical matrix contained 205 features describing baseline individual characteristics; responses to heat and cold stress; heart rate and heart-rate variability; core and skin temperature; pain tolerance; recovery; cortisol response; and environmental exposure conditions. Several physiological variables were represented by derived features such as baseline values, maxima, rates of change, between-session variability, and recovery measures. The resulting individual representation therefore integrated physiological, biochemical, individual, and environmental data within a single multimodal feature space.

2.2. Construction of the individual temperature-resilience profile

To transform the high-dimensional feature set into an interpretable individual representation, the input data were summarized into 14 components: seven describing the response to heat and seven describing the response to cold. The heat block included protocol completion, reaching the heat-tolerance limit, final core temperature, the rate of core-temperature change, heart-rate response, minimum heart-rate variability, and heart-rate-variability recovery. The cold block included cold-tolerance time, pain intensity, autonomic response, heart-rate variability and its recovery, the frequency of early withdrawal, and the number of completed cold-exposure stages.

Because the source variables were measured on different scales, component values were transformed to a common scale using robust standardization based on the median and interquartile range. Component directions were oriented so that higher values consistently represented a more favorable response to thermal stress. The seven heat components were aggregated into the Heat Resilience Index (H), and the seven cold components into the Cold Resilience Index (C).

Joint consideration of H and C produced a two-dimensional individual temperature-resilience profile in which resilience to high and low temperatures was represented separately. Four profiles were defined from the two indices: increased resilience to heat and cold, predominantly cold-resilient, predominantly heat-resilient, and reduced resilience to heat and cold. This two-dimensional representation provided an interpretable summary of individual temperature resilience for subsequent machine-learning analysis.

2.3. Machine-learning models

After the two-dimensional profile was constructed, each observation was assigned to one of the four resilience profiles using an operational classification scheme based on H and C. Three scores were calculated:

$$S_{both} = \min(H, C)$$

$$S_{heat} = H - 0.25C$$

$$S_{cold} = C - 0.25H$$

Using fixed thresholds, the increased-resilience-to-both profile was assigned first, followed by the predominantly heat-resilient and predominantly cold-resilient profiles; all remaining observations were assigned to the reduced-resilience-to-both profile. The thresholds were calibrated on the reference cohort and were not recalculated for new observations.

To evaluate how reproducibly the resulting four-profile classification could be recovered from different amounts of information, class-balanced Extra Trees models were trained under three feature-availability scenarios.

The two-index model used only H and C and assessed how well the four-profile classification could be reproduced from the compact two-dimensional representation.

The full post-protocol model used 205 features available after completion of the heat and cold protocols and assessed the separability of the four profiles in the complete multidimensional feature space.

The early-response model used the same algorithm but was restricted to 53 baseline and early-response features available before the full physiological response to thermal exposure had been observed. This model tested whether the final profile could be identified at an early stage of observation.

Comparing the three models therefore allowed us to examine how the timing and amount of available information affected the reliability of temperature-resilience profile classification.

2.4. Validation and evaluation metrics

Classification reliability was evaluated using five-fold stratified cross-validation. The dataset was partitioned into five folds; in each iteration, the model was trained on four folds and evaluated on the remaining held-out fold. This procedure was repeated five times so that every observation was evaluated once in data not used to train the corresponding model. The relative frequencies of the four temperature-resilience profiles were preserved within each fold.

Performance was assessed using several complementary metrics. Accuracy measured the overall proportion of correctly classified observations, while balanced accuracy and macro F1 accounted for the uneven representation of the four profiles. Recall was calculated separately for each profile. Because strong overall accuracy can coexist with poor recognition of rare classes, the primary criterion was the minimum recall across the four profiles; a minimum recall of at least 80% was predefined as the target performance level.

3. Results

3.1. Temperature-resilience profiles

The two-dimensional representation of individual temperature resilience yielded four profiles. The correlation between H and C was weak ($r = 0.079$), supporting their separate representation within the two-dimensional profile. The structure of the profiles and their distribution in the reference cohort are shown in Figure 1.

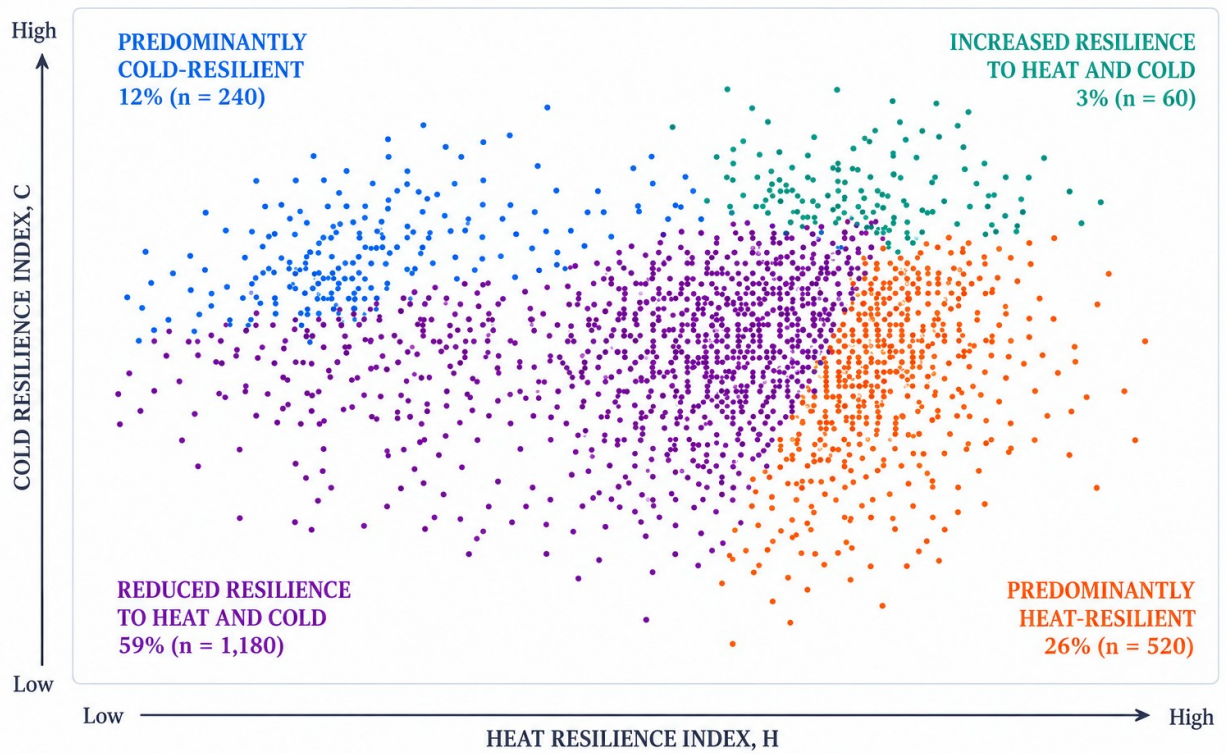


Figure 1. Temperature-resilience profiles in the H-C index space.

3.2. Machine-learning results

The performance of the three models is summarized in Table 1.

Model	Accuracy	Balanced accuracy	Macro F1	Min recall
Early-response model	63.65%	34.62%	35.26%	2.08%
Full post-protocol model	95.50%	92.65%	93.54%	86.67%
Two-index model (H, C)	98.45%	98.88%	97.91%	98.14%

Table 1. Classification performance for the four temperature-resilience profiles.

The early-response model, which used only baseline and early-response features, did not reliably identify all four profiles: its metrics remained well below the predefined target level. This indicates that early information alone was insufficient for stable identification of the final temperature-resilience profile.

The full post-protocol model achieved substantially higher classification performance and exceeded the predefined minimum-recall target of 80%. This indicates good separability of the profiles when the complete post-protocol feature set was available.

The two-index model (H, C) achieved the best overall performance among the three models. The compact two-dimensional representation therefore reproduced the four-profile classification with high accuracy.

3.3. Feature importance for temperature-resilience classification

To identify which of the 205 features were most informative for reproducing the four temperature-resilience profiles, feature importance was examined in the full post-protocol Extra Trees model. The H and C indices together accounted for approximately 48% of total feature importance, substantially more than any individual physiological feature. This indicates that the two-dimensional H-C representation concentrates a substantial portion of the information used to distinguish the four profiles (Figure 2).

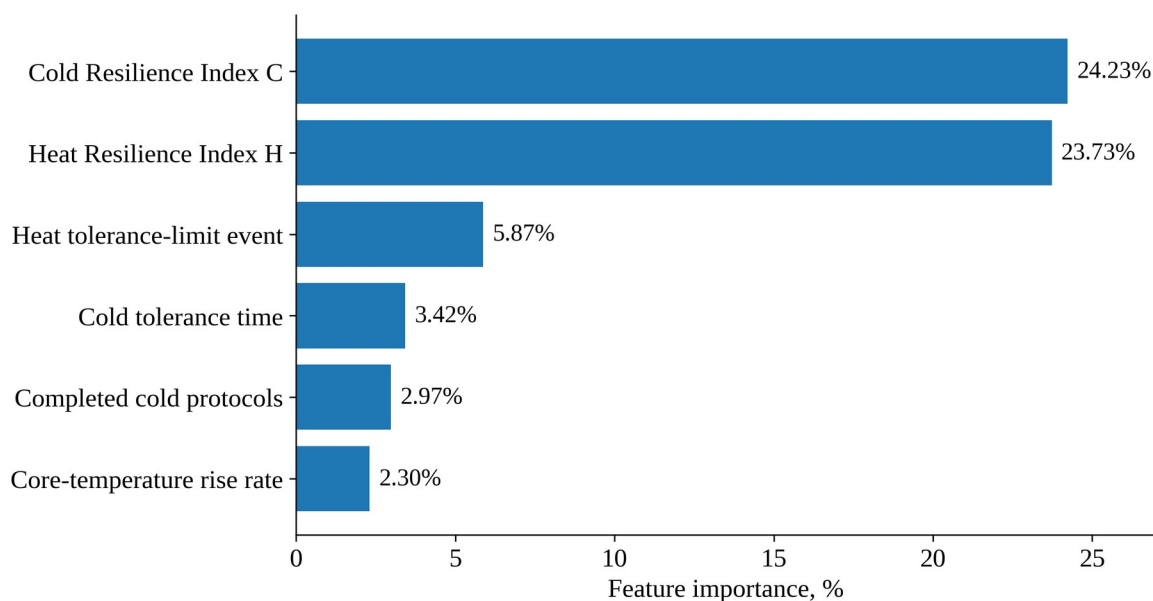


Figure 2. Feature importance in the full post-protocol Extra Trees model.

4. Discussion

The findings support the representation of temperature resilience as a two-dimensional characteristic. The weak correlation between H and C ($r = 0.079$) indicates that an individual's position on one thermal dimension is only weakly associated with their position on the other. Keeping heat and cold resilience separate therefore preserves information that could be lost if the two responses were collapsed into a single scale.

Existing studies address individual aspects of personalized thermal-response assessment but use different problem formulations and data types. Physiological heat and cold tolerance have been compared within the same participant cohort [Park et al., 2010]; four thermal-tolerance categories have been derived from subjective assessments [Kim et al., 2016]; and Kingma et al. proposed a personalized thermal-stress assessment system incorporating both personal and environmental characteristics [Kingma et al., 2021]. Machine-learning methods have also been used to estimate individual thermal tolerance in thermal-comfort applications [Zhao et al., 2026].

The present results also show that reducing the amount of available information further will require identifying a minimal feature set that preserves reliable profile classification.

The practical value of the two-dimensional profile lies in the possibility of matching an individual's combination of heat and cold resilience to the thermal demands of a specific occupation. This is relevant both

when one type of thermal exposure predominates and when high and low temperatures occur in combination. Importantly, the proposed framework characterizes an individual's temperature resilience; it does not directly determine occupational fitness. Practical deployment will require further validation in relevant occupational cohorts.

5. Conclusion

This study developed a two-dimensional personalized model of human temperature resilience that represents responses to heat and cold separately within a single individual profile.

High classification reliability was achieved when information on the individual's physiological response to thermal exposure was available.

Following further validation, the proposed framework could support standardized assessment in the selection and assignment of workers in Arctic industries, particularly in occupations involving high temperatures, low temperatures, or pronounced thermal contrasts.

6. Code and Materials Availability

Implementation code, machine-learning models, classification parameters, and computational experiment results are available in the open GitHub repository: [multimodal-ai-for-personalized-assessment-of-human-resilience-to-extreme-temperature-conditions](https://github.com/multimodal-ai-for-personalized-assessment-of-human-resilience-to-extreme-temperature-conditions).

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